

StockSense

StockSense is a smart inventory management web application built for warehouse and stock operations. It helps teams manage products, incoming stock, delivery flow, internal transfers, stock adjustments, user authentication, and profile management in one clean dashboard.

Tech Stack

- Frontend: HTML, CSS, JavaScript
- Backend: Python, Flask
- Database: Google Cloud Firestore
- Authentication: Email/password login, Google OAuth, Flask session auth
- Storage: User profile image upload support

Main Features

Inventory Management

- Product creation and update
- SKU-based product tracking
- Category management
- Unit of measure support
- Stock by warehouse or location
- Reorder level alerts

Operations

- Receipts for incoming goods
- Delivery orders for outgoing goods
- Internal stock transfers
- Stock adjustments for damaged or mismatched items
- Movement history and ledger tracking

User System

- Signup and login
- Google OAuth login
- Protected dashboard routes
- User profile page

- Custom uploaded profile picture
- Logout and session management

Dashboard UI

- Sidebar navigation
- Profile menu
- Search and filters
- Responsive tables and cards
- Attractive profile styling
- Professional business dashboard layout

Suggested Project Structure

```
StockSense/  
├── app.py  
├── config.py  
├── routes/  
│   ├── auth.py  
│   ├── profile.py  
│   ├── products.py  
│   ├── receipts.py  
│   ├── deliveries.py  
│   ├── transfers.py  
│   ├── adjustments.py  
│   └── movements.py  
├── services/  
│   └── firestore_service.py  
├── static/  
│   ├── css/  
│   │   └── style.css  
│   ├── js/  
│   │   └── app.js  
│   └── uploads/  
├── templates/  
│   ├── login.html  
│   ├── signup.html  
│   ├── index.html  
│   └── profile.html  
└── .env
```

Authentication Flow

Email/Password

1. User signs up with full name, email, password, and optional profile picture.
2. Password is hashed before storing.
3. User logs in through Flask.
4. Flask creates a session.
5. Protected dashboard becomes accessible.

Google OAuth

1. User clicks Continue with Google.
2. Google authentication flow starts.
3. Flask handles OAuth callback.
4. User profile is created or updated in Firestore.
5. Session is created and user enters the dashboard.

User Profile Image

Users can upload their own profile image instead of using a random avatar.

Recommended flow:

- validate file type
- sanitize file name
- generate unique file name
- save image in upload folder or cloud storage
- store image path/URL in Firestore
- display uploaded image in navbar, sidebar, dropdown, and profile page

Firestore Collections

Suggested collections:

users

Fields:

- id
- full_name
- email
- password_hash
- auth_provider
- google_id

- profile_picture
- role
- created_at
- updated_at

products

Fields:

- id
- name
- sku
- category
- unit
- reorder_level
- stock_by_location
- created_at
- updated_at

movements

Fields:

- id
- product_id
- movement_type
- quantity
- source_location
- destination_location
- note
- user_id
- user_name
- user_email
- created_at

Setup

1. Install dependencies

```
pip install flask google-cloud-firestore authlib python-dotenv werkzeug
```

2. Configure environment variables

Create a `.env` file:

```
FLASK_APP=app.py
FLASK_ENV=development
SECRET_KEY=your_secret_key
GOOGLE_CLIENT_ID=your_google_client_id
GOOGLE_CLIENT_SECRET=your_google_client_secret
FIRESTORE_PROJECT_ID=your_project_id
```

3. Run the app

```
flask run
```

Recommended Next Tasks

- Build profile settings page
- Add notification center
- Add role-based access control
- Add seeded demo data
- Improve settings page design
- Add final deployment polish
- Move profile image storage to Google Cloud Storage

Demo Flow

A clean demo sequence can be:

1. Signup or login
2. Enter dashboard
3. View products
4. Receive stock
5. Transfer stock
6. Deliver stock
7. Adjust inventory

8. Check movement history
9. Open profile section
10. Show uploaded profile image and user details

Notes

This project is suitable for hackathon presentation and can later be extended into a production-ready warehouse management system with stronger permissions, cloud file storage, analytics, and reporting.